



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 15

Assignment: OOAD Case Study and CASE Tool Implementation

PROGRAMME	BSc Information Communication and Technology
COURSE CODE / YEAR	BICT 3202 · Year 3
LECTURER	Francis Fweta — ICT Lecturer
DELIVERY	2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 15 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 15 module and your lecture notes before attempting each question.
- Use the university course registration case study for your examples.
- Show that your models are consistent: messages must match operations.
- Draw UML diagrams in correct notation, or with a CASE/UML tool.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define a CASE tool and state three ways it supports software development. [3 marks]

Question A2

Differentiate functional from non-functional requirements, with one example each. [3 marks]

Question A3

Define an actor and explain why an actor need not be a human being. [2 marks]

Question A4

Explain the relationship between requirements and use cases. [2 marks]

Question A5

What is model integration and why is it important? [2 marks]

Question A6

Define traceability and explain its value in OOAD. [2 marks]

Question A7

Explain how CASE tools support model consistency. [2 marks]

Question A8

State three benefits of using CASE tools in OOAD. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Consider the university course registration system as a complete case study.

- (a) Integrate all models (object, dynamic, behavioural) into a complete design. [4 marks]
- (b) Show how the models are consistent with each other. [4 marks]
- (c) Explain how CASE tools could support this integration. [4 marks]

Question B2 (8 marks)

Consider implementing the registration system with a CASE tool.

- (a) Identify three CASE tool features that would be valuable. [4 marks]
- (b) Explain how these features support the development process. [4 marks]

Section C — Case Study (10 marks)

Consider a complete OOAD project for a banking system.

As an object-oriented analyst/designer:

- (a) Explain how you would use CASE tools throughout the project lifecycle. [4 marks]
- (b) Identify potential challenges and how CASE tools help address them. [3 marks]
- (c) Explain the value of model integration for project success. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — banking system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Ensure your models are complete and consistent.
- Cite any sources you consult using the format used in the Week 15 module references.